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Roll No

EC-3002-CBGS

B.E. III Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Digital Circuits and Systems

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Implement Ex-OR and Ex-NOR gate using
 - i) NAND gate
 - ii) NOR gate
- b) Implement the following Boolean functions using all NAND gates
 - i) $\overline{A}BC + \overline{B}(\overline{C+D}) + \overline{AB} \cdot \overline{CD}$
 - ii) $B(\overline{CD} + AC) + \overline{ABC} D + \overline{BC}(D + \overline{D})$
2. Express the following Boolean functions in SOP and POS form.
 - i) $\overline{A}(B + C + \overline{D}) + \overline{AC} + \overline{B}(C + D) + \overline{BC}$
 - ii) $AC + BD + \overline{AD}(B + C)$
3. Minimize the expression using Quine McCluskey method
$$Y = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + A\overline{B}\overline{C}\overline{D} + A\overline{B}\overline{C}D + A\overline{B}C\overline{D} + A\overline{B}C\overline{D}$$

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4.
 - a) Design a carry look ahead adder.
 - b) Design a magnitude comparator
5.
 - a) Design a 3 bit Binary to Gray code converter and draw logic diagram for it.
 - b) Draw logic diagram, truth table and excitation table of JK Flip Flop. Write its merits over SR FF.
6.
 - a) What is D Flip Flop? Draw logic diagram, truth table and excitation table of JK Flip Flop. How to realize this using SR Flip Flop?
 - b) What is MOD counter? Design a MOD 6 counter using D Flip Flop.
7.
 - a) Compare ROM, PAL and PLA.
 - b) Describe the methods of storing data in ROM, PROM and EPROM. What are their relative merits?
8. Write short notes on any two:
 - a) PLA
 - b) ECL family
 - c) Ring counter
 - d) Exclusive OR gate
